Institutional Quantitative Architecture Specification
FINRIX CLI — XAUUSD Forecasting and Execution Platform

Academic Project Report | Dual-Path Quantitative Research and Trading System

Abstract

FINRIX is an institutional-style quantitative terminal for XAUUSD research, forecasting, strategy replay, risk sizing, and execution planning. The platform combines a local CPU path with a Modal A10G Cloud GPU path, allowing the same validated data, forecasting, technical-indicator, and risk abstractions to run interactively from the finrix command-line interface. Its design emphasizes lookahead-safe validation, explicit numerical guards, configurable indicators, and transparent output artifacts rather than opaque trading decisions.

The system includes regularized Temporal Fusion Transformer forecasting, log-return target scaling, quantile corridors, mean-reverting Monte Carlo volatility envelopes, dynamic equity-based position sizing, and a 14-indicator technical strategy suite. This report documents the architecture, mathematical foundations, validation controls, security posture, and planned extensions.

1. System Scope and Design Principles

- Market: XAUUSD and gold-linked macro drivers, including DXY, Treasury yields, GVZ, ETF flows, CFTC positioning, and filtered macro news.
- Interface: FINRIX interactive terminal with Rich rendering, persistent settings, configurable indicator parameters, and direct forecast execution.
- Execution modes: Local CPU for deterministic offline or low-cost analysis; Modal A10G Cloud GPU for NeuralForecast TFT and parallel strategy workloads.
- Validation: Lookahead-safe historical replay, quantile ordering checks, finite-value checks, dynamic risk sensitivity tests, and artifact verification.
- Operational invariant: forecasts and strategy results are advisory research outputs; live execution requires an explicit broker or MT5 bridge.

2. Dual-Path System Architecture

FINRIX routes work according to data/finrix_config.json. The same account equity, risk percentage, and indicator overrides are loaded before Option 7 and Option 8 operations. This prevents the user interface from silently using stale defaults.

Layer	Local CPU Path	Modal A10G Cloud GPU Path
Launcher	python -m macro_forecast_engine.run_engine	python -m modal run macro_forecast_engine/modal_engine.py
Market ingestion	Yahoo Finance with ticker fallback and local cache	Yahoo Finance inside the Modal image with the same fallback logic
Forecasting	Calibrated local trend and forecast worker	NeuralForecast TFT with PyTorch Lightning
Strategy replay	trader_timemachine.py on local data	Parallel strategy_backtest_worker.map execution
Artifacts	data/macro_*_forecast.json	data/macro_*_forecast.json and

		modal_ensemble.json
Primary use	Fast iteration, offline checks, and interactive exploration	GPU forecasting, parallel ensembles, and institutional-scale runs

Both paths use UTF-8-safe subprocess handling, explicit error propagation, and the same validated parameter schema. Modal execution additionally passes the indicator parameter payload to workers so remote jobs cannot silently revert to standard settings.

3. Deep Learning and Quantile Mathematics

3.1 Temporal Fusion Transformer

The TFT path models log returns rather than nominal prices. For price P_t, the training target is $y_t = \ln(P_t / P_{t-1})$. Forecast returns are compounded back to price levels after prediction. This improves numerical conditioning across long horizons and prevents the model from learning a largely monotonic price scale.

The model uses multi-quantile loss. For quantile level τ and residual $u = y - \hat{y}\tau$, the pinball loss is:

$$L_\tau(y, \hat{y}\tau) = \max(\tau u, (\tau - 1)u) = \tau \cdot \max(u, 0) + (1 - \tau) \cdot \max(-u, 0).$$

FINRIX estimates $\tau \in \{0.10, 0.50, 0.90\}$. The required corridor invariant is $q10(t) \le q50(t) \le q90(t)$ for every forecast row. The TFT configuration uses dropout = 0.30, AdamW weight decay = 1e−4, validation monitoring, and early stopping patience of 15 validation checks.

3.2 Dynamic Volatility Envelope

Rather than flattening q10 at a static price floor, FINRIX uses a smooth trailing log-normal envelope around the median path. With median price m_t, trailing volatility σ_t, historic drift μ, and $z = 1.2816$ for the 10th/90th percentile:

$$q10(t) = m_t \times \exp(-z \times \Sigma_t), \quad q90(t) = m_t \times \exp(+z \times \Sigma_t)$$

$$\Sigma_t = \mathrm{sqrt}(\Sigma_{i=1..t} \sigma_i^2)$$

The trailing volatility path is exponentially damped toward a long-run estimate: $\sigma_t = \sigma_{long} + (\sigma_{current} - \sigma_{long}) \exp(-t / 30)$. An annualized volatility budget limits numerical explosion while preserving row-to-row movement. Monte Carlo paths use mean-reverting deviations $d_t = \rho d_{t-1} + \varepsilon_t$ with $\rho = 0.92$.

4. Institutional Risk Management

Risk is calculated from current equity at every simulated trade rather than from static percentage returns. This makes the backtester sensitive to both account equity and risk-per-trade settings.

Quantity	Formula
Dynamic risk dollars	$R_t = Equity_t \times (risk_per_trade_pct / 100)$
Stop distance	$StopPips_t = \lvert Entry_t - StopLoss_t \rvert / PipSize$
Lot size	$Lots_t = R_t / (StopPips_t \times PipValue)$
Trade P&L	$PnL_t = Lots_t \times PipChange_t \times PipValue$
Equity update	$Equity_{t+1} = \max(Equity_t + PnL_t, \varepsilon)$
Net return	$100 \times (FinalEquity - StartingEquity) / StartingEquity$
Drawdown	$100 \times \min(Equity_t / PeakEquity_t - 1)$ in absolute magnitude

A zero-division epsilon and minimum positive equity guard prevent invalid sizing. The execution layer calculates structural SMC stops and multi-stage targets, while the risk layer applies fractional Kelly, volatility parity, and the broker micro-lot floor.

5. Fourteen-Indicator Technical Strategy Suite

Strategy	Method
SMC Fair Value Gap (FVG)	Detects price displacement gaps against a configurable minimum pip gap.
SMC Liquidity Order Block Breakout	Tests breaks beyond rolling liquidity highs and lows.
EMA Golden Cross / Death Cross	Compares configurable fast, slow, and trend EMAs.
Triple EMA Ribbon	Classifies alignment of fast, medium, and slow exponential averages.
RSI Mean Reversion	Uses configurable RSI period, overbought, and oversold thresholds.
MACD Zero-Line & Signal Line Crossover	Combines MACD line, signal line, and zero-line direction.
Bollinger Bands Squeeze & Reversal	Identifies deviations from a configurable rolling mean and band width.
Supertrend ATR Trend Follower	Uses ATR-period volatility and a configurable multiplier.
Stochastic Oscillator	Compares K and D lines over configurable lookback and slowing periods.
ADX + DI Trend Classifier	Filters directional movement by configurable ADX strength.
Ichimoku Cloud Kumo Breakout	Classifies price relative to Tenkan, Kijun, and Senkou cloud values.
ATR Volatility Expansion Channel	Detects price movement beyond an ATR-based exponential channel.
TFT Log-Return Trend	Uses rolling log-return drift as the technical replay proxy for the ML trend.
Volatility Parity & Risk-Adjusted Allocation	Scales directional exposure by trailing realized volatility.

5.1 Persistent Indicator Overrides

The file data/finrix_indicator_params.json stores validated MT5/TradingView-style overrides. Options 7 and 8 load this file before calculation and offer an interactive standard-versus-custom parameter menu. RSI periods are constrained to 2–100; all other periods, thresholds, factors, and SMC lookbacks are validated before saving.

Group	Parameters
RSI	period, overbought, oversold
EMA	fast_period, slow_period, trend_period
MACD	fast, slow, signal
Bollinger	period, std_dev
Supertrend	atr_period, factor
Stochastic	k_period, d_period, slowing
ADX / ATR	period, threshold
Ichimoku	tenkan, kijun, senkou_b
SMC	fvg_min_pip_gap, order_block_lookback

6. Backtesting and Evaluation Protocol

Each signal is shifted by one observation before P&L is calculated. This prevents the current bar from influencing a trade entered on that same bar. Option 7 supports multi-selection across 15m, 1h, 4h, and 1D data. Option 8 runs the same matrix locally or in parallel on Modal A10G and ranks the top five strategies.

- Win Rate: percentage of simulated trades with positive P&L.
- Profit Factor: gross winning dollars divided by absolute gross losing dollars.
- Maximum Drawdown: dollar equity peak-to-trough loss expressed as a percentage.
- Sharpe Ratio: annualized mean strategy return divided by return volatility.
- Net Expected Return: compounded final equity relative to starting equity.
- Sensitivity: lowering risk from 2% to 1% changes dollar P&L, return, and drawdown; changing RSI 14 to RSI 28 changes trigger timing and metrics.

7. Zero-Trust Security and Reliability Audit

The verification program recorded 145+ tests PASSED across compilation, local and Modal routing, strategy selection, parameter validation, forecast artifacts, risk sensitivity, quantile ordering, and Windows UTF-8 execution. The audit posture treats every external feed, model output, and cached artifact as untrusted input.

- Finite-value guards reject or normalize inf and NaN values before sizing and serialization.
- Zero-division epsilon protects equity, volatility, stop-distance, and profit-factor calculations.
- Quantile ordering is enforced after fallback and TFT prediction paths.
- Yahoo Finance intraday limits are normalized to 60d for 15m/1h/4h and 730d for 1D.
- Gold ingestion falls back from GC=F to XAUUSD=X and then to a local cache.
- Subprocesses use UTF-8 encoding and replacement errors on Windows.
- Modal cloud-credit execution requires an explicit user confirmation for the ensemble.
- No lookahead is permitted in replay signals or forecast origins.

These controls reduce operational failure modes but do not eliminate market risk, model risk, data vendor outages, slippage, liquidity risk, or the possibility of adverse regime change.

8. Future Roadmap

1. Extend the feature and indicator schema to major Forex pairs with pair-specific pip sizes, session calendars, and liquidity profiles.
2. Add Crypto market adapters with 24/7 calendar handling, exchange normalization, and venue-aware fee models.
3. Implement a direct MT5 execution bridge with explicit order previews, broker-symbol validation, two-step confirmation, and kill-switch controls.
4. Add portfolio-level correlation, exposure, and margin constraints across multiple instruments.
5. Introduce reproducible experiment manifests, model registry metadata, and scheduled drift monitoring.
6. Expand the institutional audit suite with property-based tests and deterministic historical snapshots.

Conclusion

FINRIX provides a coherent research-to-execution foundation: the local and cloud paths share validated data contracts, forecasts are expressed as calibrated quantile corridors, backtests use dynamic equity compounding, and technical strategies are configurable rather than hard-coded. Its next stage is controlled multi-asset expansion and a guarded broker bridge, preserving the zero-trust and lookahead-safe principles documented here.